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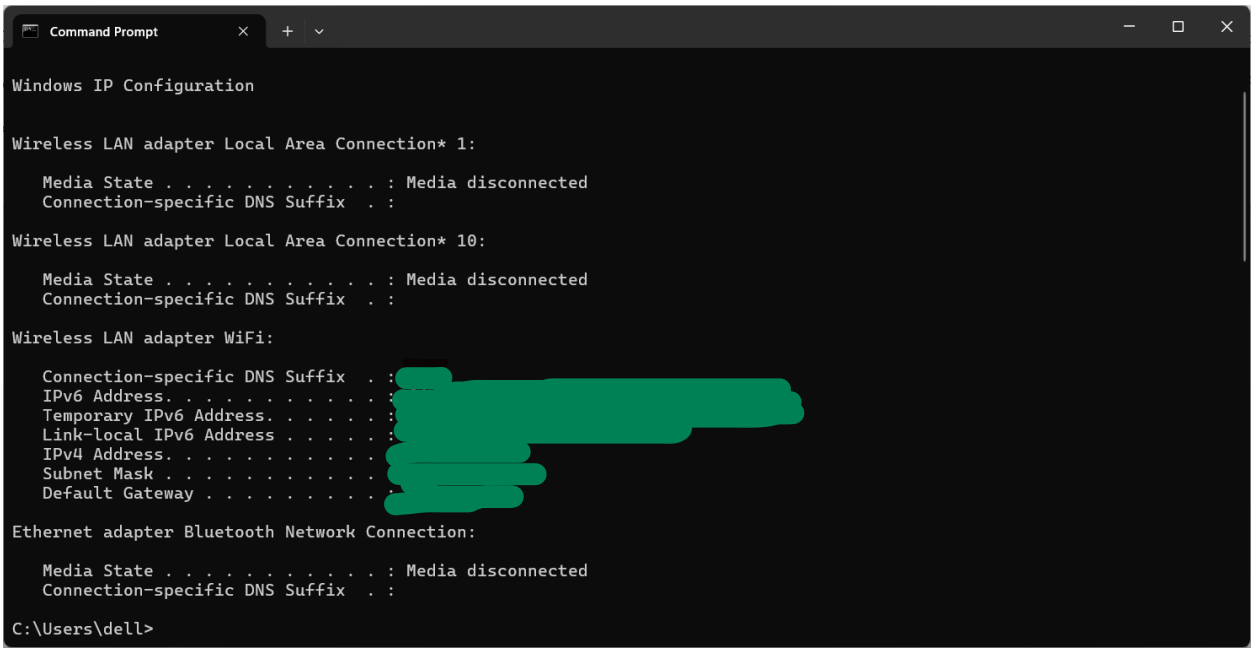
Lap_Network_WorkShop

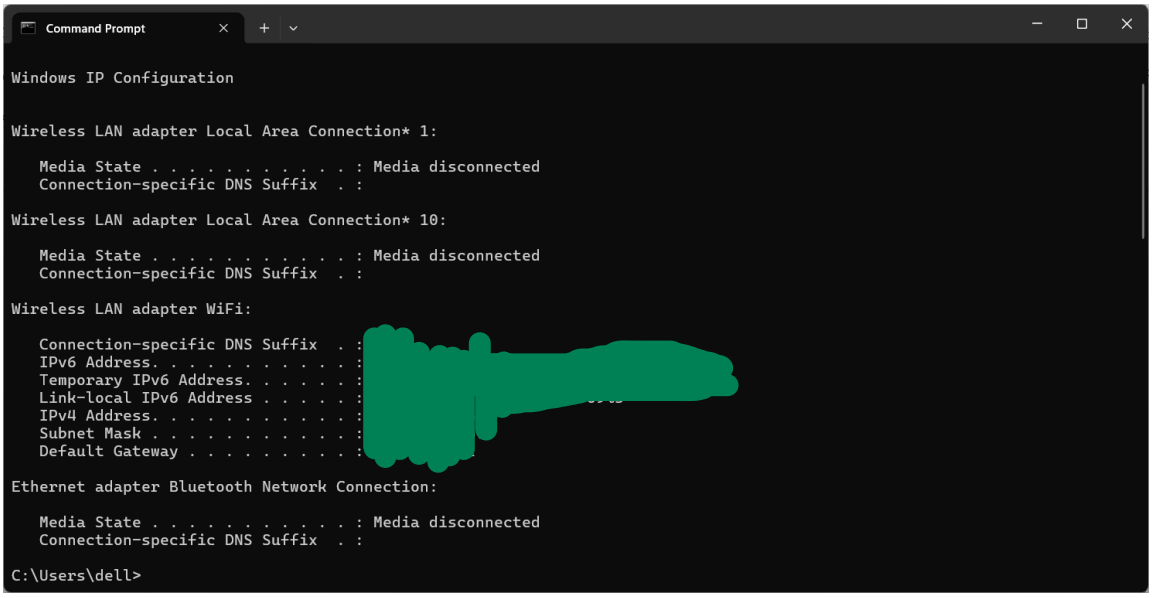
Q1: explain how to get the private IP?

- Using **ipconfig** command in Command Prompt
- From **Network settings** in Control Panel



Q2: What is the difference between ipconfig & ipconfig/all?

- Ipconfig:



- Ipconfig/all

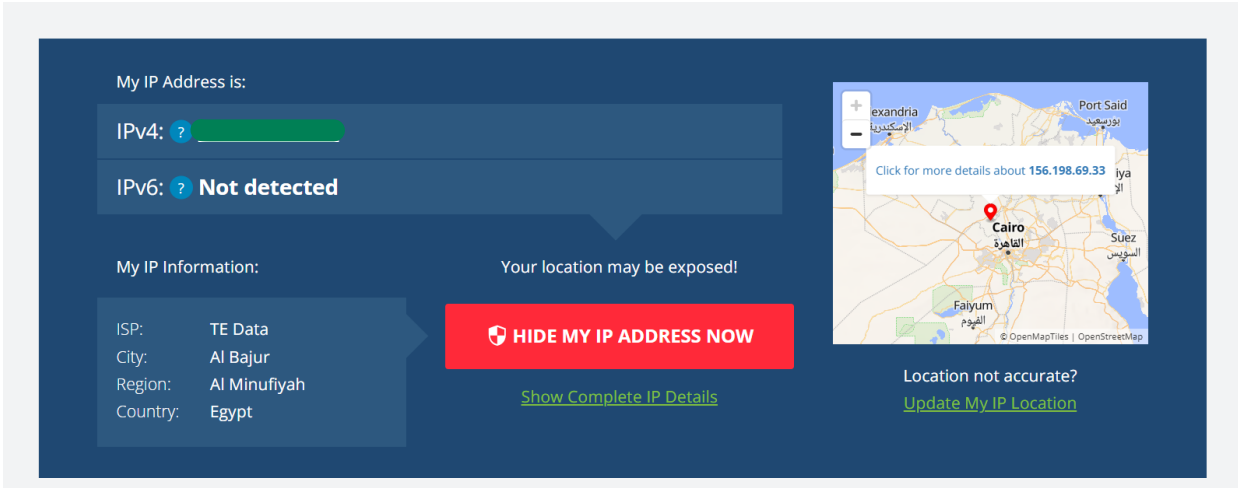
```
Command Prompt

C:\Users\dell>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-RQ52ECS
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : home

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : [redacted]
   Physical Address. . . . . . . . . : [redacted]
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : [redacted]
   Physical Address. . . . . . . . . : [redacted]
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter WiFi:
```

-

```
Command Prompt

   Connection-specific DNS Suffix  . : [redacted]
   Description . . . . . . . . . . . :
   Physical Address. . . . . . . . . :
   DHCP Enabled. . . . . . . . . . . :
   Autoconfiguration Enabled . . . . :
   IPv6 Address. . . . . . . . . . . :
   Temporary IPv6 Address. . . . . . :
   Link-local IPv6 Address . . . . . :
   IPv4 Address. . . . . . . . . . . :
   Subnet Mask . . . . . . . . . . . :
   Lease Obtained. . . . . . . . . . :
   Lease Expires . . . . . . . . . . :
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . :
   DHCPv6 Client DUID. . . . . . . . :
   DNS Servers . . . . . . . . . . . :

   NetBIOS over Tcpip. . . . . . . . : [redacted]

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 0C-DD-24-27-2D-05
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

ipconfig displays basic network configuration, while ipconfig /all displays detailed information including MAC address, DHCP, and DNS servers.

Q3: explain how to get the public IP?



- Visit a website like “what is my IP”
- Or use router settings
- Or command: `nslookup myip.opendns.com resolver1.opendns.com`

Q4: What is the difference between public ip & private ip?

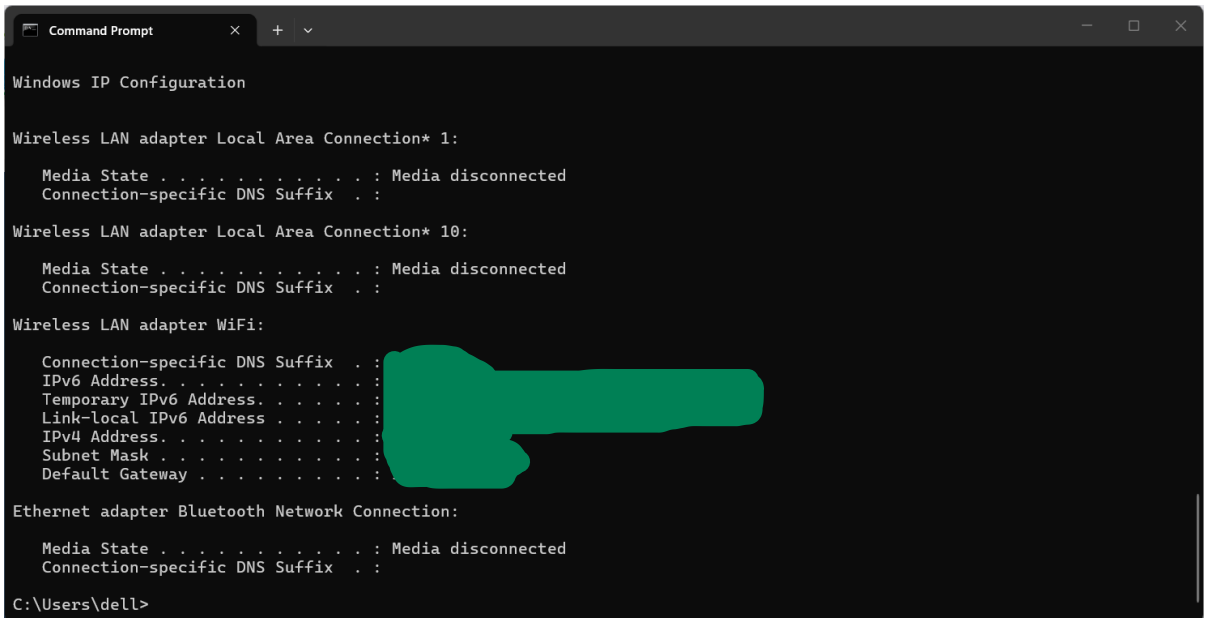
Public IP Address

- A **Public IP** is assigned by the **Internet Service Provider (ISP)**.
- It is **unique globally**.
- Used to identify a device or network on the **Internet**.
- Accessible from anywhere on the Internet.

```
C:\Users\dell>nslookup myip.opendns.com resolver1.opendns.com
DNS request timed out.
    timeout was 2 seconds.
Server:  UnKnown
Address:  208.67.222.222

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
*** Request to UnKnown timed-out

C:\Users\dell>
```

Private IP Address

- A **Private IP** is used inside a **local network (LAN)**.
- Not reachable from the Internet.
- Assigned by a **DHCP server** or manually.

- Used for communication inside the local network.
- 

Q5: How does the device get its IP?

Automatically (DHCP)

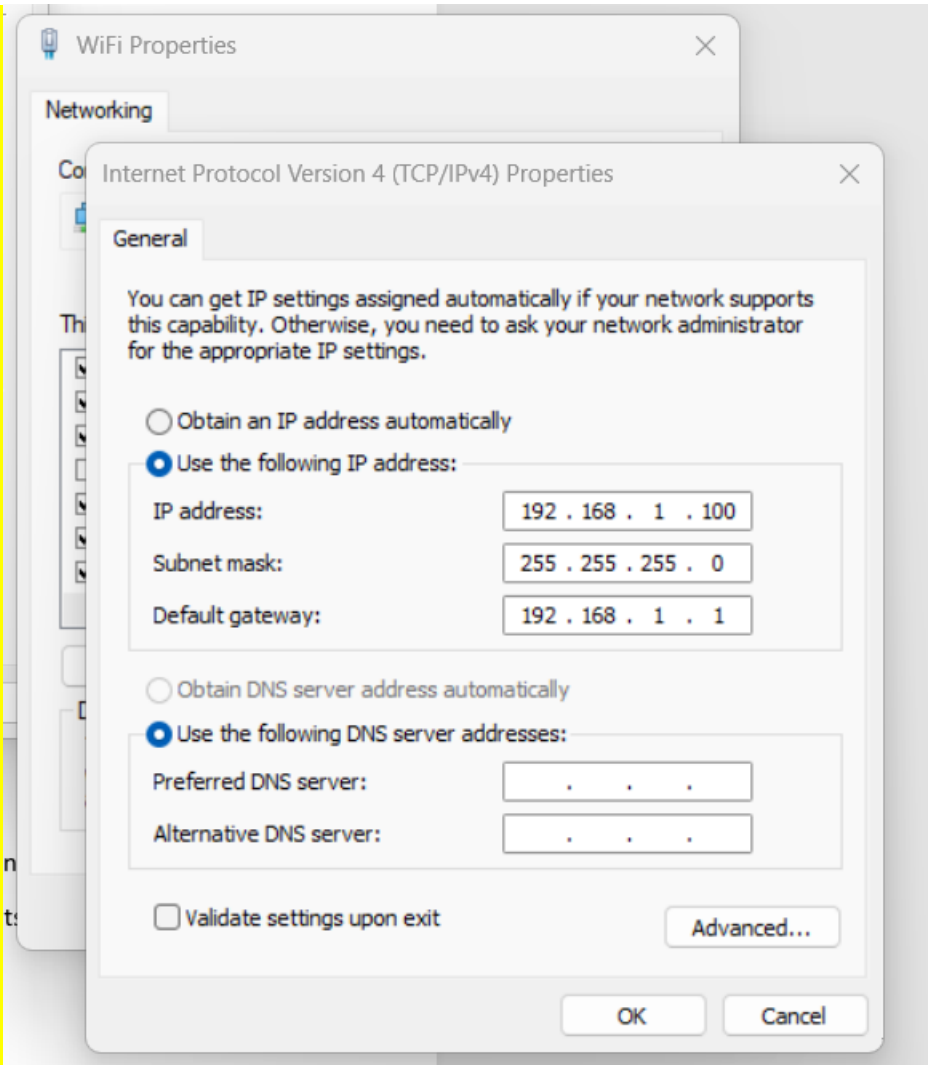
- The device gets its IP address automatically from a **DHCP Server**.
- The DHCP server is usually the **router**.
- This process is called **DHCP lease**.

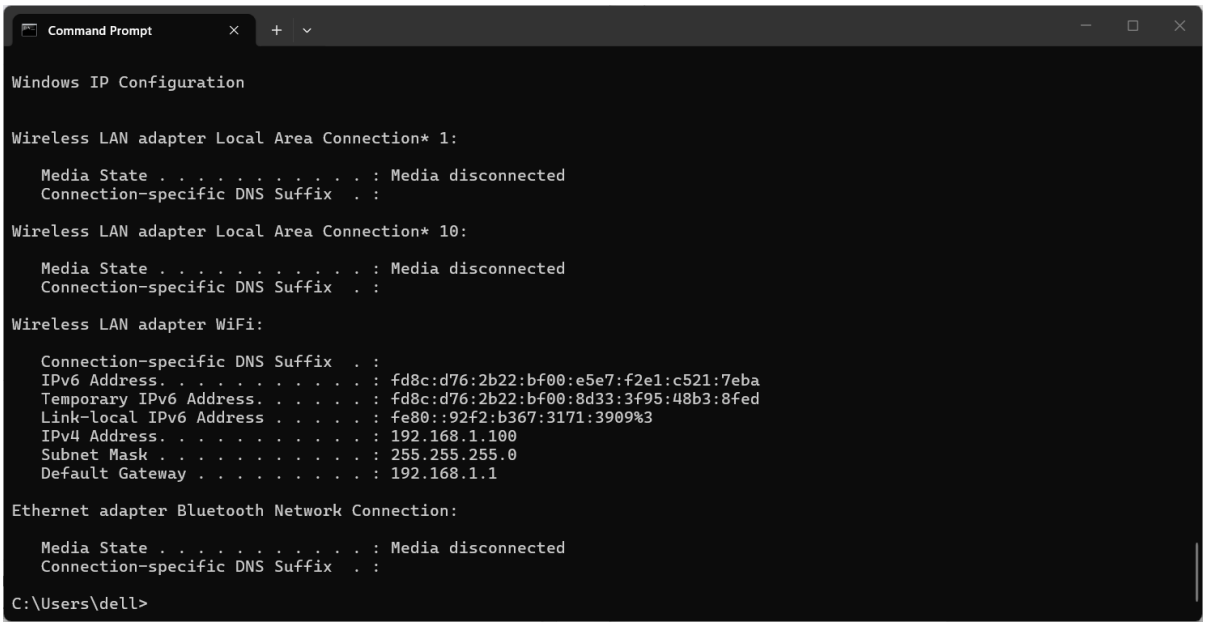
Q6: Make your device get its private IP statically?

1. Open **Control Panel**
2. Go to **Network and Internet → Network and Sharing Center**
3. Click on your active network → **Properties**
4. Select **Internet Protocol Version 4 (TCP/IPv4)** → **Properties**
5. Select **Use the following IP address**
6. Enter:
    a. **IP Address:** 192.168.1.100 (example)
    b. **Subnet Mask:** 255.255.255.0
    c. **Default Gateway:** 192.168.1.1
7. Click **OK** → Close all windows
8. Open **Command Prompt**
9. Type:

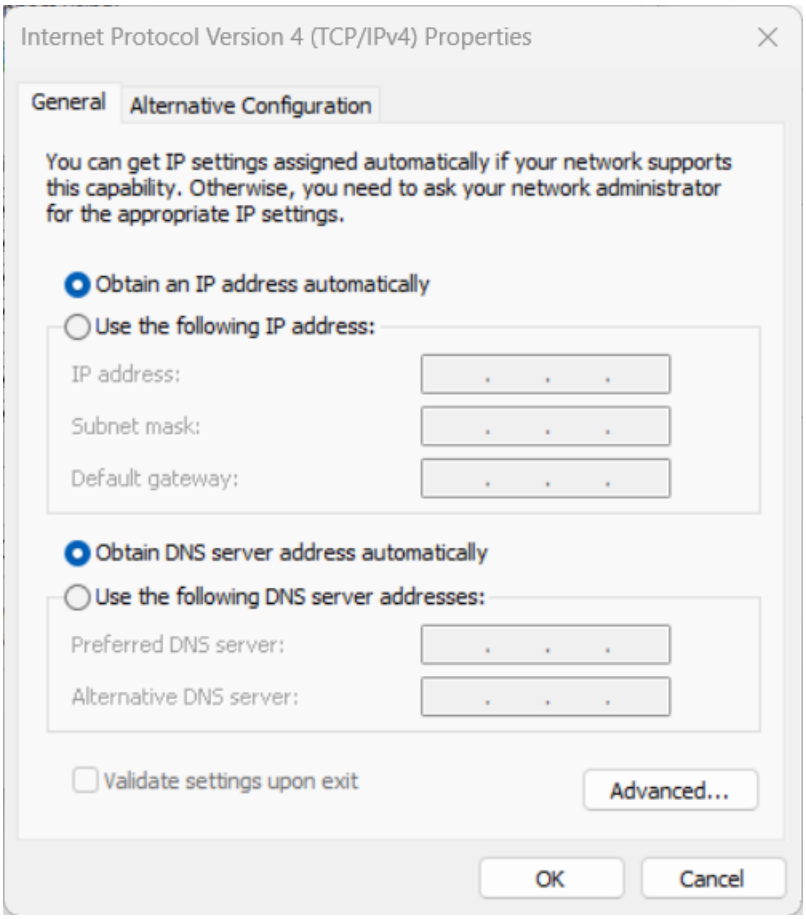
```
ipconfig
```

10. Press **Enter** and verify your **IPv4 Address** changed to the static IP.





Q7: Reset your device to get its private IP automatically?



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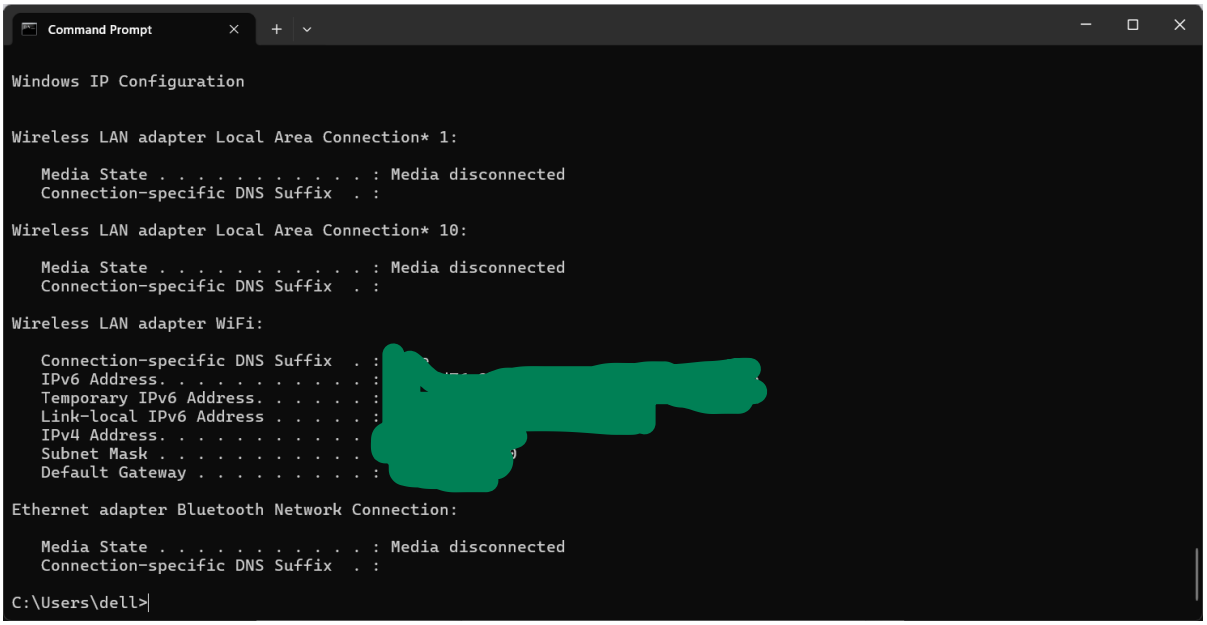


Q8: What do you know about APIPA Address ?

APIPA stands for **Automatic Private IP Addressing**.

- When a device cannot get an IP address from a DHCP server, it automatically assigns itself an IP in the range:

`169.254.0.1 – 169.254.255.254`

- This IP is used **temporarily** so the device can communicate within the local network only.
- The device **cannot access the internet** until it gets a valid IP from a DHCP server.
- APIPA is useful for small networks or troubleshooting when DHCP is unavailable.

How to check APIPA:

1. Disconnect from the network or disable DHCP temporarily.
2. Open Command Prompt.
3. Type `ipconfig` and press Enter.
4. Look for IPv4 address starting with **169.254.x.x**

Q9: Give me 2 ways to find out your device's MAC address.

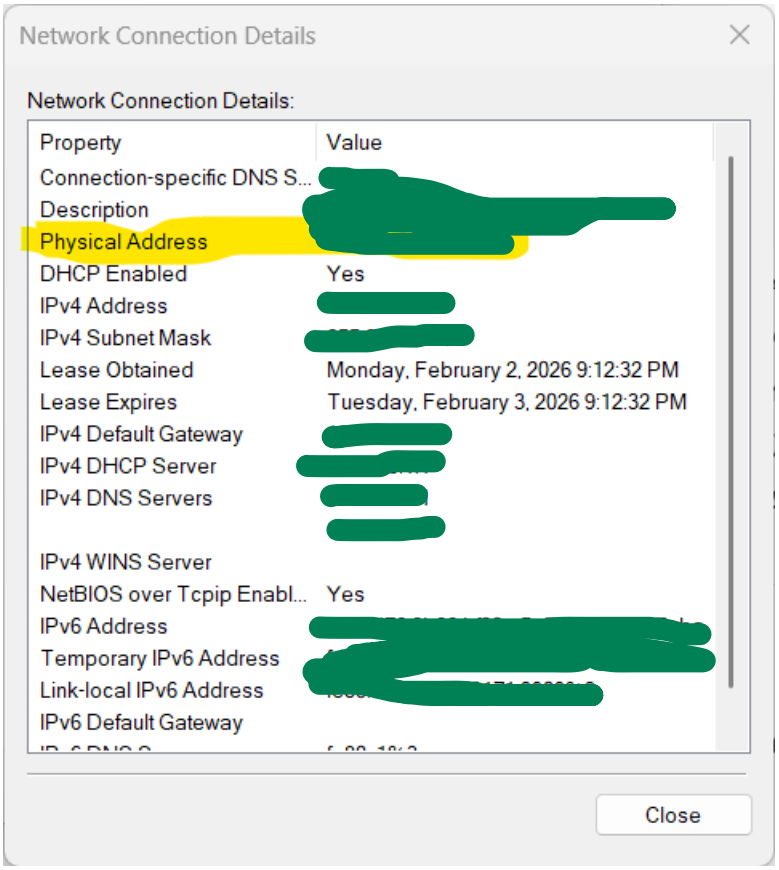
1. getmac

```
C:\Users\dell>getmac

Physical Address    Transport Name
=================== ==========================================================
[redacted]          \Device\Tcpip_{1C834FEF-CB4B-4381-8FF7-F063FF5329B9}
[redacted]          Media disconnected

C:\Users\dell>
```

2. Control Panel



Q10: What is the difference between getmac & getmac/v ?

Command	Description	Output
`getmac`	Shows only the **MAC address** of each network adapter	Simple list of MAC addresses
`getmac /v`	Shows **verbose details** about each network adapter	Includes: MAC address, adapter name, connection name, and status

```
C:\Users\dell>getmac

Physical Address    Transport Name
=================== ==========================================================
0C-DD-24-27-2D-01   \Device\Tcpip_{1C834FEF-CB4B-4381-8FF7-F063FF5329B9}
0C-DD-24-27-2D-05   Media disconnected

C:\Users\dell>getmac /v

Connection Name Network Adapter Physical Address    Transport Name
=============== =============== =================== ==========================================================
WiFi            Intel(R) Wirele [redacted]          \Device\Tcpip_[redacted]
Bluetooth Netwo Bluetooth Devic [redacted]          Media disconnected

C:\Users\dell>
```

Q16. Can you tell me which command that could check connectivity between 2 devices? And then:

Command to check connectivity: `ping <IP_Address>`

Ping Command:

- Used to check network connectivity between devices.
- Sends small packets to the target IP and waits for replies.

a) check connectivity of your loopback IP address “127.0.0.1” ?

```
C:\Users\dell>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\dell>
```

b) How many packets are sent to check availability?

Answer: 4 packets (on Windows)

c) How many packages did your device receive? What does this mean?

- 4 received → NIC working fine, no packet loss
- Fewer than 4 → some issue in NIC, drivers, or firewall

d) Check connectivity between your device and this IP "10.10.0.10". then explain the result?

-
  - **Reply from 10.10.0.10** → device reachable, network OK
  - **Request timed out** → device unreachable, network problem, or firewall blocking

```
C:\Users\dell>ping 10.10.0.10

Pinging 10.10.0.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\dell>
```

Q17. Verify the connectivity of the loopback IP address "127.0.0.1" by sending “8” packets which the size of each packet is "50000".?

```
C:\Users\dell>ping 127.0.0.1 -n 8 -l 50000

Pinging 127.0.0.1 with 50000 bytes of data:
Reply from 127.0.0.1: bytes=50000 time<1ms TTL=128
Reply from 127.0.0.1: bytes=50000 time<1ms TTL=128
Reply from 127.0.0.1: bytes=50000 time<1ms TTL=128
Reply from 127.0.0.1: bytes=50000 time<1ms TTL=128
Reply from 127.0.0.1: bytes=50000 time<1ms TTL=128
Reply from 127.0.0.1: bytes=50000 time<1ms TTL=128
Reply from 127.0.0.1: bytes=50000 time<1ms TTL=128
Reply from 127.0.0.1: bytes=50000 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 8, Received = 8, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\dell>
```

Q18. Explain what is the meaning of this command: “ping 127.0.0.1 -t “؟

1. **`ping`** – This is a network utility used to check if a host (computer or device) is reachable over the network. It also measures the round-trip time for messages sent to that host.
2. **`127.0.0.1`** – This is the **loopback IP address**, also called **localhost**. It refers to your **own computer**. Pinging this address tests whether your computer’s network stack (TCP/IP) is working properly, without sending data to the outside network.
3. **`-t`** – This option tells `ping` to **keep sending ping requests continuously** until you manually stop it (usually by pressing `Ctrl + C`). Without `-t`, `ping` sends only 4 requests by default on Windows.

So, the full command means:

Continuously send ping requests to your own computer (localhost) to test whether the network stack is working and monitor response times.

Use case:

- Checking if your TCP/IP stack is working.
- Monitoring your network for any sudden failures or delays.

```
C:\Users\dell>ping 127.0.0.1 -t

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
^C
C:\Users\dell>
```

Q19. What is the meaning of " DOS Attack" ?

DOS Attack stands for **Denial of Service Attack**.

- It is a type of **cyber attack** where an attacker tries to make a computer, server, or network **unavailable to its users**.
- The attacker does this by **overloading the target with too many requests** or consuming its resources (like memory, CPU, or network bandwidth), so legitimate users **cannot access the services**.

Key points:

- Target: website, server, or network.
- Goal: disrupt normal operation.
- Effect: slow performance or complete unavailability.

Example:

- Sending thousands of requests per second to a website until it crashes